# COMSATS UNIVERSITY ISLAMABAD,WAH CAMPUS

## CIVIL ENGINEERING

### SUBJECT: PROPERTIES OF CONCRETE

**PBL / ASSIGNMENT**

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# ABSTRACT

This report investigates the properties of concrete using natural aggregates (NA) and recycled aggregates (RA). The study evaluates the density, moisture absorption, compressive strength, and UPV (Ultrasonic Pulse Velocity) of concrete samples, both with natural aggregates and a mix incorporating 50% recycled aggregates. The results show a reduction in compressive strength and dry density with the use of recycled aggregates, largely due to the poor quality of cement and improper sample preparation. Additionally, slump tests indicated minimal workability, highlighting the need for consistent material properties. Recommendations are provided for improving future mix designs and testing procedures.


# INTRODUCTION AND LITERATURE REVIEW

## INTRODUCTION

The use of recycled aggregates (RAs) in concrete production represents a sustainable solution for the construction industry, addressing both environmental and economic concerns. By substituting natural aggregates (NAs) with RAs derived from construction and demolition waste, we reduce the demand for virgin resources and the environmental impact of concrete production. This substitution supports sustainability by reducing landfill waste, conserving natural resources, and potentially lowering transportation and material costs. However, incorporating RAs into concrete also introduces challenges, particularly related to variability in RA quality, which can influence the consistency and performance of concrete.

The use of recycled aggregates (RA) from treated construction and demolition waste (CDW) as replacement of natural aggregates (NA) in the production of new construction products has been considered one of the most efficient methods to add value to these materials, frequently regarded as worthless. Indeed, as an alternative to depositing them in landfills, the use of RA creates new market opportunities, which are also favorable to the environment. **[1]**

## THE IMPACT OF RA CONTENT ON CONCRETE PROPERTIES

### 1. COMPRESSIVE STRENGTH:

Studies indicate that RA content affects the compressive strength of concrete. For example, mixes with RA replacement levels of 20% or less show only slight reductions in compressive strength compared to conventional concrete. At higher replacement levels, such as 50% or 100%, the reduction in strength becomes more significant. This is often due to the porous nature of RAs, which tend to absorb more water, leading to weaker interfacial transition zones (ITZ) between aggregates and cement paste. A study by Poon et al. (2004) **[2]** noted that concrete with 50% RA exhibited around 10–15% lower compressive strength than conventional concrete, while Silva et al. (2015) found that 100% RA content could result in a 25–30% reduction in compressive strength.

## 2. SLUMP AND WORKABILITY:

Increased RA content generally results in reduced workability due to the higher water absorption of RAs. Concrete mixes with high RA levels often show lower slump values, indicating decreased fluidity and ease of handling. To counteract this effect, additional water or admixtures may be needed, which can influence the final properties of the concrete. According to Silva et al. (2015) **[1]**, a 50% RA mix exhibited a slump reduction of around 20–25% compared to an NA mix.

It has been found that the workability of concrete with natural and recycled aggregate is almost the same if water saturated surface dry recycled aggregate is used. Also, if dried recycled aggregate is used and additional water quantity is added during mixing, the same workability can be achieved after a prescribed time. Bulk density of fresh concrete is slightly decreased with an increase in the quantity of recycled aggregate. **[3]**

## 3. EFFECT OF WATER-TO-CEMENT RATIO

The water-to-cement (w/c) ratio is crucial in determining concrete properties, especially with RAs. For instance, a low w/c ratio (e.g., 0.4) can help maintain compressive strength in mixes with moderate RA content, as it limits the amount of excess water that can be absorbed by the aggregates. However, a higher w/c ratio (e.g., 0.6) may be needed to improve workability in high-RA mixes. According to Yang et al. (2008) **[4]**, concrete with a 0.5 w/c ratio and 50% RA content achieved satisfactory compressive strength but required careful mix adjustments for optimum workability.

It was stated that the reduction in compressive strength of recycled concrete was largely controlled by water/cement ratio, water absorption of RCA and replacement ratio. Rao et al. **[5]** found that the strength of concrete with RCA and NA was comparable even at full replacement, but the water/cement ratio needed to be maintained at a ratio higher than 0.55. If the water/cement ratio was reduced to 0.4, the strength of RCA was only 75% control mix.

## PROPERTIES OF RECYCLED AGGREGATES VS. NATURAL AGGREGATES

RAs differ from NAs in several keyways:

- Water Absorption: RAs generally have higher water absorption due to residual cement paste and micro-cracks from previous use, which can affect concrete's water demand and workability.
- Density: RAs often have a lower density than NAs, potentially affecting the weight of the concrete mix and resulting in slightly lower strength.
- Particle Size: RAs may contain irregular particles with variable shapes and sizes, which can reduce the ease of compaction and finishing in concrete applications.

## TESTS ON CONCRETE WITH RECYCLED AGGREGATES

**Table 1:-**

| SR.NO. | PARTICULARS | VALUES | |
|---|---|---|---|
| | | Natural Aggregate | Recycled Coarse Aggregate |
| 1 | Specific Gravity | 2.4-3.0 | 2.35-2.58 |
| 2 | Water Absorption | 0.29%-0.3% | 0.3%-0.32% |
| 3 | Bulk Density | 1678.2 KN/$m^3$ | 1469.8KN/$m^3$ |
| 4 | Crushing Values: | 18.4% | 36.3% |
| 5 | Impact Values: | 17.65% | 35.2% |

**Table 2: Compressive strength**

| Compressive Strength | Replacement Of Natural Aggregate | | | |
|---|---|---|---|---|
| | 0% | 10% | 20% | 30% |
| M30-3Days | 20.63 N/$mm^2$ | 16.38 N/$mm^2$ | 19.05 N/$mm^2$ | 18.46 N/$mm^2$ |
| M30-7 Days | 33.13 N/$mm^2$ | 23.83 N/$mm^2$ | 31.93 N/$mm^2$ | 28.05 N/$mm^2$ |
| M30-28 Days | 47.53 N/$mm^2$ | 42.28 N/$mm^2$ | 43.92 N/$mm^2$ | 40.27 N/$mm^2$ |
| M40-3 Days | 31.59 N/$mm^2$ | 28.44 N/$mm^2$ | 27.56 N/$mm^2$ | 25.78 N/$mm^2$ |
| M40-7 Days | 56.67 N/$mm^2$ | 53.69 N/$mm^2$ | 51.69 N/$mm^2$ | 49.78 N/$mm^2$ |
| M40-28 Days | 64.42 N/$mm^2$ | 60.44 N/$mm^2$ | 56.22 N/$mm^2$ | 54.22 N/$mm^2$ |

**Table 3: Flexural Strength**

| Flexural Strength | Replacement Of Natural Aggregate | | | |
|---|---|---|---|---|
| | 0% | 10% | 20% | 30% |
| M30-7 Days | 3.58 N/$mm^2$ | 3.04 N/$mm^2$ | 3.52 N/$mm^2$ | 3.30 N/$mm^2$ |
| M30-28 Days | 4.98 N/$mm^2$ | 4.71 N/$mm^2$ | 4.805 N/$mm^2$ | 4.601 N/$mm^2$ |
| M40-7 Days | 4.69 N/$mm^2$ | 4.57 N/$mm^2$ | 4.48 N/$mm^2$ | 4.396 N/$mm^2$ |
| M40-28 Days | 5.818 N/$mm^2$ | 5.637 N/$mm^2$ | 5.436 N/$mm^2$ | 5.334 N/$mm^2$ |

*Mr. Tushar R Sonawane, Prof. Dr. Sunil S. Pimplikar, Use of Recycled Aggregate Concrete, IOSR Journal of Mechanical and Civil Engineering (IOSR-JMCE)*

The use of recycled aggregates presents a viable pathway toward sustainable construction. Key findings indicate that while RAs can replace NAs to some extent, especially at lower levels, challenges related to strength, workability, and durability persist. Many studies highlight the importance of controlling the w/c ratio and understanding the unique properties of RAs to achieve consistent performance. However, gaps remain in the literature regarding the long-term durability of high-RA-content concrete under varying environmental conditions. This study will address these gaps by systematically examining RA content and w/c ratio impacts on concrete properties, contributing to a more robust understanding of recycled aggregate use in sustainable construction.

## MATERIALS

The materials used in this experiment include natural aggregates (NA), recycled aggregates (RA), cement, and water. The key physical properties of NA and RA are summarized below.

**Natural Aggregates**

The natural aggregates were provided by the university. Their properties, including density and absorption, are listed in the table below:

| NATURAL AGGREGATES | |
|---|---|
| Oven Dry weight (g) → A | 2500 |
| SSD or Surface Dry Weight (g) → B | 2545 |
| Weight of Wet Agg (g) → C | 1427 |
| Absorption Capacity % | 1.8 |
| Specific Gravity | 2.28 |

**Recycled Aggregates**

The recycled aggregates were obtained by breaking old concrete debris sourced from the university. The properties of the recycled aggregates are as follows:

| RECYCLED AGGREGATES | |
|---|---|
| Oven Dry weight (g) → A | 1433 |
| SSD or Surface Dry Weight (g) → B | 1510 |
| Weight of Wet Agg (g) → C | 845 |
| Absorption Capacity % | 5.37 |
| Specific Gravity | 2.27 |

| Materials | Dry Density (kg/m³) |
|---|---|
| Natural Aggregate | 1396.96 |
| Recycle Aggregate | 1210.00 |





# MIX DESIGN

For this experiment, two concrete mix designs were prepared:

1. **Control Mix**: Using natural aggregates.
2. **Assigned Mix**: Incorporating 50% recycled aggregates.

The following parameters were used:

- Mix ratio: **1:1:2** (Cement: Fine Aggregate: Coarse Aggregate)
- Water-Cement Ratio (w/c): **0.4**
- Specimen Size: **4 inches × 8 inches cylinders**
- Number of Molds: **3 for NA samples, 3 for RA samples**

The volumes were calculated as follows:

- **Volume for 1 Cylinder ($mm^3$):** 1,570,796 $mm^3$
- **Total Volume for 3 Cylinders:** 4,712,388.98 $mm^3$
- **Accounting for Wastage (15%) + Wet to Dry Adjustment (52%):** Final Total Volume = **0.01647 $m^3$**

**Mix Design Formula:**

The concrete mix design was calculated using the following equations:

1. Volume of Concrete = Weight / Density
2. Mass of Cement = (Volume of Cement / Total Volume) × Total Weight
3. Water Required = Water-Cement Ratio × Cement Weight

The control mix and assigned mix used the same water-cement ratio (0.4), but the assigned mix replaced 50% of the natural coarse aggregates with recycled aggregates.

# METHODOLOGY

The following steps outline the testing procedures:

**1. Material Preparation:**

- Natural aggregates were dried and weighed to determine oven-dry and SSD weights.
- Recycled aggregates were sourced, crushed, and dried.

**2. Mix Preparation:**

- Batching of materials was conducted based on the mix ratio of 1:1:2.

- Water was added to achieve a 0.4 water-cement ratio.
- Mixing was performed manually to ensure homogeneity.

**3. Slump Test:**

- The slump test was performed following ASTM C143 standards.
- The test measured workability and slump of both mixes.

**4. Mold Preparation:**

- Cylindrical molds (4" x 8") were filled with the concrete mix in three layers.
- Each layer was compacted to minimize air voids.

**5. Curing:**

- The specimens were demolded after 24 hours and cured for 14 days.

**6. Testing:**

- **Compressive Strength Test:** Conducted using a UTM (Universal Testing Machine).
- **UPV Test:** Conducted per ASTM C597 to determine concrete quality and homogeneity.





## RESULTS

The results for slump, compressive strength, UPV, and dry density are summarized in the tables below:

| | Control Mix | | | |
|---|---|---|---|---|
| | **Slump (mm)** | **Dry density (kg/m3)** | **UPV (km/s)** | **Comp Strength (Mpa)** |
| **SAMPLE 1** | 0 | 2335.21 | 2.48 | 7.86 |
| **SAMPLE 2** | 0 | 2365.54 | 2.45 | 17.43 |
| **SAMPLE 3** | 0 | 2244.23 | 2.43 | 15.81 |

| | Assigned Mix | | | |
|---|---|---|---|---|
| | **Slump (mm)** | **Dry density (kg/m3)** | **UPV (km/s)** | **Comp Strength (Mpa)** |
| **SAMPLE 1** | 0 | 2292.75 | 3.19 | 16.8 |
| **SAMPLE 2** | 0 | 2323.08 | 2.55 | 15.51 |
| **SAMPLE 3** | 0 | 2244.23 | 2.54 | 16.02 |







## DISCUSSION AND CONCLUSION

The results indicate a reduction in compressive strength and dry density when using 50% recycled aggregates. Key observations include:

1. **Compressive Strength:** The control mix achieved an average of **13.7 MPa**, whereas the assigned mix achieved **16.11 MPa**. However, one sample of the control mix (Sample 1) performed poorly (7.86 MPa) due to improper preparation.
2. **Slump Results:** The zero slump in both mixes indicates poor workability, likely caused by inconsistent or poor-quality cement and insufficient moisture in the aggregates.
3. **Dry Density:** A slight decrease in density for the assigned mix reflects the use of recycled aggregates, which have lower specific gravity and higher absorption.
4. **UPV Results:** The UPV readings show slightly better values for the assigned mix, indicating acceptable concrete quality.

The lower strength and workability can be attributed to factors such as poor cement quality, improper mixing, and sample preparation issues.

## RECOMMENDATIONS

1. Use **fresh cement** to ensure consistent strength and quality.
2. Properly monitor the **moisture content** of aggregates before mixing.
3. Ensure correct **compaction** of concrete to avoid air voids.
4. Conduct tests after a standard **28-day curing period** for reliable results.
5. Use higher-quality recycled aggregates with proper treatment to improve performance.

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